Problem 3

Part a)

Part a) i): $X_nY_n \xrightarrow{d} cX$

For every $n \in \mathbb{N}$, we have that:

$$\begin{aligned} X_nY_n &= X_n(Y_n + c - c) \\ &= X_nY_n + cX_n - cX_n \\ &= cX_n + X_n(Y_n - c), \end{aligned} \tag{1}$$

which holds pointwise for every $\omega \in \Omega$. We will show that $cX_n \xrightarrow{d} cX$, $X_n(Y_n - c) \xrightarrow{\mathbb{P}} 0$, and then combine the two results to show $X_nY_n \xrightarrow{d} cX$.

Step 1 (Show $\forall\delta > 0\ \exists M > 0 : \mathbb{P}(|X_n| > M) < \delta, \forall n \in \mathbb{N}$). Let $\delta > 0$ be arbitrarily fixed, and let $D := \{x \in \mathbb{R} : F_X(x) = F_X(x-)\}$ the set of continuity points of F_X, as in Theorem 4.1.6. Since F_X is a distribution function on $\mathbb{R}$, we know $\lim_{x\to-\infty} F_X(x) = 0$ and $\lim_{x\to\infty} F_X(x) = 1$ (Lemma 1.4.2c). Hence we may choose $a', b' \in \mathbb{R}$ with

$$F_X(a') < \frac{\delta}{4}, \qquad F_X(b') > 1 - \frac{\delta}{4}.$$

A distribution function has at most countably many discontinuity points, so D^c is at most countable. Therefore, since F_X is monotonically increasing (Lemma 1.4.2b), we may replace a' by some $a \le a'$ with $a \in D$ and replace b' by some $b \ge b'$ with $b \in D$, and still have:

$$F_X(a) \le F_X(a') < \frac{\delta}{4}, \qquad F_X(b) \ge F_X(b') > 1 - \frac{\delta}{4}.$$

Since $X_n \xrightarrow{d} X$, Theorem 4.1.6 gives $F_{X_n}(a) \to F_X(a)$ and $F_{X_n}(b) \to F_X(b)$ as $n \to \infty$. Applying the definition of this limit, there exists $N \in \mathbb{N}$ such that for all $n \ge N$,

$$|F_{X_n}(a) - F_X(a)| < \frac{\delta}{4}, \qquad |F_{X_n}(b) - F_X(b)| < \frac{\delta}{4}.$$

In particular, for all $n \ge N$,

$$F_{X_n}(a) = F_X(a) + \big(F_{X_n}(a) - F_X(a)\big) \ \le\ F_X(a) + |F_{X_n}(a) - F_X(a)| \ <\ \frac{\delta}{4} + \frac{\delta}{4} = \frac{\delta}{2},$$
$$1 - F_{X_n}(b) = \big(1 - F_X(b)\big) + \big(F_X(b) - F_{X_n}(b)\big) \ \le\ \big(1 - F_X(b)\big) + |F_{X_n}(b) - F_X(b)| \ <\ \frac{\delta}{4} + \frac{\delta}{4} = \frac{\delta}{2},$$

i.e. $F_{X_n}(a) < \delta/2$ and $F_{X_n}(b) > 1 - \delta/2$. Setting $M_0 := \max(|a|, |b|)$, we have:

$$\{|X_n| > M_0\} = \{X_n < -M_0\} \cup \{X_n > M_0\} \subseteq \{X_n \le a\} \cup \{X_n > b\}$$

Therefore, by subadditivity (Lemma 1.1.10e):

$$\mathbb{P}(|X_n| > M_0) \le \mathbb{P}(X_n \le a) + \mathbb{P}(X_n > b) = F_{X_n}(a) + \big(1 - F_{X_n}(b)\big) < \frac{\delta}{2} + \frac{\delta}{2} = \delta, \quad \forall n \ge N.$$

For the finitely many remaining indices $1, \dots, N-1$, fix $n \in \{1, \dots, N-1\}$. The sequence of events $(\{|X_n| > M\})_{M\in\mathbb{N}}$ is monotonically decreasing in M with $\bigcap_{M\in\mathbb{N}}\{|X_n| > M\} = \emptyset$ (since $X_n(\omega) \in \mathbb{R}$ is finite for every $\omega \in \Omega$, so no ω can satisfy $|X_n(\omega)| > M$ for all $M \in \mathbb{N}$ simultaneously). By Corollary 1.1.16a (continuity of $\mathbb{P}$ from above):

$$\mathbb{P}(|X_n| > M) \xrightarrow{M\to\infty} \mathbb{P}(\emptyset) = 0,$$

Therefore, for each fixed $n \in \{1, \dots, N-1\}$:

$$\exists M_n > 0 : \quad \mathbb{P}(|X_n| > M_n) < \delta$$

Now set $M := \max(M_0, M_1, \dots, M_{N-1})$. We check the two cases:

- **Case $n \geq N$:** Since $M \geq M_0$, monotonicity of $\mathbb{P}$ (Lemma 1.1.10d) applied to $\{|X_n| > M\} \subseteq \{|X_n| > M_0\}$ gives:

$$\mathbb{P}(|X_n| > M) \leq \mathbb{P}(|X_n| > M_0) < \delta.$$

- **Case $n \in \{1, \dots, N-1\}$:** Since $M \geq M_n$, monotonicity of $\mathbb{P}$ applied to $\{|X_n| > M\} \subseteq \{|X_n| > M_n\}$ gives

$$\mathbb{P}(|X_n| > M) \leq \mathbb{P}(|X_n| > M_n) < \delta.$$

In both cases $\mathbb{P}(|X_n| > M) < \delta$. Thus, $\forall \delta > 0 \; \exists M > 0 : \mathbb{P}(|X_n| > M) < \delta, \forall n \in \mathbb{N}$.

Step 2 ($X_n(Y_n - c) \xrightarrow{\mathbb{P}} 0$). Let $\epsilon > 0$ and let $\delta > 0$ be arbitrary. Also, choose $M > 0$ such that $\mathbb{P}(|X_n| > M) < \delta$ for all $n \in \mathbb{N}$ (which exists by Step 1). From Step 1, we have $|X_n(\omega)| \leq M$, then for any $\omega \in \Omega$,

$$|X_n(\omega)(Y_n(\omega) - c)| = |X_n(\omega)| \cdot |Y_n(\omega) - c| \;\leq\; M \cdot |Y_n(\omega) - c|$$

Then:

$$M \cdot |Y_n(\omega) - c| \leq \varepsilon \qquad \Longleftrightarrow \qquad |Y_n(\omega) - c| \leq \frac{\varepsilon}{M}$$

Therefore, whenever $\omega \in \Omega$ satisfies $|X_n(\omega)| \leq M$ and $|Y_n(\omega) - c| \leq \varepsilon/M$, we get:

$$|X_n(\omega)(Y_n(\omega) - c)| \;\leq\; M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

In set notation, this is equivalent to:

$$\begin{aligned}
&\{|X_n| \leq M\} \cap \left\{|Y_n - c| \leq \frac{\varepsilon}{M}\right\} \;\subseteq\; \{|X_n(Y_n - c)| \leq \varepsilon\} \\
\Longrightarrow\; &\{|X_n(Y_n - c)| \leq \varepsilon\}^c \subseteq \left(\{|X_n| \leq M\} \cap \left\{|Y_n - c| \leq \frac{\varepsilon}{M}\right\}\right)^c && \text{(taking complements)} \\
\Longrightarrow\; &\{|X_n(Y_n - c)| > \varepsilon\} \subseteq \{|X_n| > M\} \cup \left\{|Y_n - c| > \frac{\varepsilon}{M}\right\} && \text{(De Morgan's law)}
\end{aligned}$$

By sub-additivity (Lemma 1.1.10e), we have:

$$\begin{aligned}
\mathbb{P}(|X_n(Y_n - c)| > \varepsilon) \leq\; & \mathbb{P}(|X_n| > M) + \mathbb{P}\Big(|Y_n - c| > \frac{\varepsilon}{M}\Big) \\
<\; & \delta + \mathbb{P}\Big(|Y_n - c| > \frac{\varepsilon}{M}\Big) \quad \text{(by Step 1)}
\end{aligned}$$

Taking $\limsup_{n\to\infty}$ on both sides and using $Y_n \xrightarrow{\mathbb{P}} c$, we get:

$$\begin{aligned}
\limsup_{n\to\infty} \mathbb{P}(|X_n(Y_n - c)| > \varepsilon) \;&\leq\; \limsup_{n\to\infty} \Big[\delta + \mathbb{P}\Big(|Y_n - c| > \frac{\varepsilon}{M}\Big)\Big] \\
&=\; \delta + \limsup_{n\to\infty} \mathbb{P}\Big(|Y_n - c| > \frac{\varepsilon}{M}\Big) \\
&=\; \delta + \lim_{n\to\infty} \mathbb{P}\Big(|Y_n - c| > \frac{\varepsilon}{M}\Big) && \text{(since the limit exists)} \\
&=\; \delta + 0 && \text{(since } Y_n \xrightarrow{\mathbb{P}} c\text{, Definition 4.1.1)} \\
&=\; \delta.
\end{aligned}$$

Since $\delta > 0$ was arbitrary, letting $\delta \downarrow 0$ gives

$$\begin{aligned}
&\limsup_{n\to\infty} \mathbb{P}(|X_n(Y_n - c)| > \varepsilon) \le 0 \\
\implies &\limsup_{n\to\infty} \mathbb{P}(|X_n(Y_n - c)| > \varepsilon) = 0 \quad (\mathbb{P} \ge 0)
\end{aligned}$$

As $\varepsilon > 0$ was arbitrary, Definition 4.1.1 gives:

$$X_n(Y_n - c) \xrightarrow{\mathbb{P}} 0.$$

Step 3 ($cX_n \xrightarrow{d} cX$). Let $h : \mathbb{R} \to \mathbb{R}$ be bounded and continuous, and $c \in \mathbb{R}$ an arbitrary constant. Define $g(x) := h(cx)$ for $x \in \mathbb{R}$. Then g is again bounded (since h is bounded) and continuous (as a composition of the continuous map $x \mapsto cx$ with the continuous map h). Since $X_n \xrightarrow{d} X$ (Definition 4.1.4(ii)), then:

$$\mathbb{E}[g(X_n)] \xrightarrow{n\to\infty} \mathbb{E}[g(X)],$$

which is equivalent to,

$$\mathbb{E}[h(cX_n)] \xrightarrow{n\to\infty} \mathbb{E}[h(cX)].$$

As h was an arbitrary bounded continuous function, then, by Definition 4.1.4(ii), we have:

$$cX_n \xrightarrow{d} cX.$$

Step 4 ($A_n \xrightarrow{d} A,\ B_n \xrightarrow{\mathbb{P}} 0 \implies A_n + B_n \xrightarrow{d} A$). Let $(A_n)_{n\in\mathbb{N}}$, A, and $(B_n)_{n\in\mathbb{N}}$ be real-valued random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Assume that $A_n \xrightarrow{d} A$ and $B_n \xrightarrow{\mathbb{P}} 0$. Let $x \in D_A := \{y \in \mathbb{R} : F_A(y) = F_A(y-)\}$ (the set of continuity points of F_A, as in Theorem 4.1.6) be arbitrarily fixed, and let $\varepsilon > 0$ be arbitrary. First, we will find upper and lower bounds for $F_{A_n+B_n}(x)$:

- **Upper bound.** For any $\omega \in \Omega$ with $A_n(\omega) > x + \varepsilon$ and $|B_n(\omega)| \le \varepsilon$ (i.e. $B_n(\omega) \ge -\varepsilon$), we get:

  $$A_n(\omega) + B_n(\omega) \ > \ (x + \varepsilon) + (-\varepsilon) \ = \ x.$$

  In set notation, this is equivalent to:

  $$\begin{aligned}
  &\{A_n > x + \varepsilon\} \cap \{|B_n| \le \varepsilon\} \ \subseteq \ \{A_n + B_n > x\} \\
  \implies &\{A_n + B_n > x\}^c \ \subseteq \ \big(\{A_n > x + \varepsilon\} \cap \{|B_n| \le \varepsilon\}\big)^c \qquad \text{(taking complements)} \\
  \implies &\{A_n + B_n \le x\} \ \subseteq \ \{A_n \le x + \varepsilon\} \cup \{|B_n| > \varepsilon\} \qquad \text{(De Morgan's law)}
  \end{aligned}$$

  By subadditivity (Lemma 1.1.10e), we have:

  $$\mathbb{P}(A_n + B_n \le x) \ \le \ \mathbb{P}(A_n \le x + \varepsilon) + \mathbb{P}(|B_n| > \varepsilon),$$

  i.e.,

  $$F_{A_n+B_n}(x) \ \le \ F_{A_n}(x + \varepsilon) + \mathbb{P}(|B_n| > \varepsilon) \tag{2}$$

- **Lower bound.** For any $\omega \in \Omega$ with $A_n(\omega) \le x - \varepsilon$ and $|B_n(\omega)| \le \varepsilon$ (i.e. $B_n(\omega) \le \varepsilon$), we get:

$$A_n(\omega) + B_n(\omega) \;\le\; (x - \varepsilon) + \varepsilon \;=\; x.$$

Therefore, whenever $\omega \in \Omega$ satisfies $A_n(\omega) \le x - \varepsilon$ and $|B_n(\omega)| \le \varepsilon$, we get $A_n(\omega) + B_n(\omega) \le x$. In set notation, this is equivalent to:

$$\begin{aligned}
& \{A_n \le x - \varepsilon\} \cap \{|B_n| \le \varepsilon\} \;\subseteq\; \{A_n + B_n \le x\} & \\
\implies\; & \{A_n + B_n \le x\}^c \;\subseteq\; \big(\{A_n \le x - \varepsilon\} \cap \{|B_n| \le \varepsilon\}\big)^c & \text{(taking complements)} \\
\implies\; & \{A_n + B_n > x\} \;\subseteq\; \{A_n > x - \varepsilon\} \cup \{|B_n| > \varepsilon\} & \text{(De Morgan's law)}
\end{aligned}$$

By subadditivity (Lemma 1.1.10e), we have:

$$\mathbb{P}(A_n + B_n > x) \;\le\; \mathbb{P}(A_n > x - \varepsilon) + \mathbb{P}(|B_n| > \varepsilon),$$

i.e.,

$$1 - F_{A_n+B_n}(x) \;\le\; \big(1 - F_{A_n}(x - \varepsilon)\big) + \mathbb{P}(|B_n| > \varepsilon),$$

which, after rearranging (subtract 1 from both sides, multiply by -1), gives:

$$F_{A_n}(x - \varepsilon) \;\le\; F_{A_n+B_n}(x) + \mathbb{P}(|B_n| > \varepsilon) \tag{3}$$

Combining (2) and (3). For every $n \in \mathbb{N}$ and every $\varepsilon > 0$:

$$F_{A_n}(x - \varepsilon) - \mathbb{P}(|B_n| > \varepsilon) \;\le\; F_{A_n+B_n}(x) \;\le\; F_{A_n}(x + \varepsilon) + \mathbb{P}(|B_n| > \varepsilon). \tag{4}$$

Since D_A^c is at most countable (a distribution function has at most countably many discontinuity points), we may choose a sequence $\varepsilon_k \downarrow 0$ such that $x - \varepsilon_k \in D_A$ and $x + \varepsilon_k \in D_A$ for every $k \in \mathbb{N}$. Fix $k \in \mathbb{N}$ and let $n \to \infty$ in (4) with $\varepsilon = \varepsilon_k$. Since $A_n \xrightarrow{d} A$ and $x \pm \varepsilon_k \in D_A$, Theorem 4.1.6 gives $F_{A_n}(x \pm \varepsilon_k) \to F_A(x \pm \varepsilon_k)$ as $n \to \infty$. Since $B_n \xrightarrow{\mathbb{P}} 0$, Definition 4.1.1 gives $\mathbb{P}(|B_n| > \varepsilon_k) \to 0$ as $n \to \infty$. Taking $\liminf_n$ on the left part of (4) and $\limsup_n$ on the right part:

$$\begin{aligned}
\liminf_{n\to\infty} F_{A_n+B_n}(x) \;&\ge\; \liminf_{n\to\infty} \Big[F_{A_n}(x - \varepsilon_k) - \mathbb{P}(|B_n| > \varepsilon_k)\Big] \;=\; F_A(x - \varepsilon_k) - 0 \;=\; F_A(x - \varepsilon_k), \\
\limsup_{n\to\infty} F_{A_n+B_n}(x) \;&\le\; \limsup_{n\to\infty} \Big[F_{A_n}(x + \varepsilon_k) + \mathbb{P}(|B_n| > \varepsilon_k)\Big] \;=\; F_A(x + \varepsilon_k) + 0 \;=\; F_A(x + \varepsilon_k)
\end{aligned}$$

Combining these two inequalities:

$$F_A(x - \varepsilon_k) \;\le\; \liminf_{n\to\infty} F_{A_n+B_n}(x) \;\le\; \limsup_{n\to\infty} F_{A_n+B_n}(x) \;\le\; F_A(x + \varepsilon_k) \tag{5}$$

Now let $k \to \infty$ in (5), i.e. $\varepsilon_k \downarrow 0$:

$$\begin{aligned}
F_A(x + \varepsilon_k) \;&\xrightarrow{k\to\infty}\; F_A(x+) \;=\; F_A(x) && (F_A \text{ right-continuous, Lemma 1.4.2d}) \\
F_A(x - \varepsilon_k) \;&\xrightarrow{k\to\infty}\; F_A(x-) \;=\; F_A(x) && (\text{since } x \in D_A, \text{ i.e. } F_A(x) = F_A(x-))
\end{aligned}$$

Plugging these two limits into (5) gives:

$$F_A(x) \;\le\; \liminf_{n\to\infty} F_{A_n+B_n}(x) \;\le\; \limsup_{n\to\infty} F_{A_n+B_n}(x) \;\le\; F_A(x),$$

which forces:

$$\lim_{n\to\infty} F_{A_n+B_n}(x) \;=\; F_A(x).$$

As $x \in D_A$ was arbitrary, Theorem 4.1.6 gives:

$$A_n + B_n \xrightarrow{d} A.$$

Step 5 ($X_nY_n \xrightarrow{d} cX$). Define $A_n := cX_n$ and $B_n := X_n(Y_n - c)$ for $n \in \mathbb{N}$. We check that the two hypotheses of Step 4 are satisfied for this choice:

- By Step 3, $A_n = cX_n \xrightarrow{d} cX$. We therefore set $A := cX$.
- By Step 2, $B_n = X_n(Y_n - c) \xrightarrow{\mathbb{P}} 0$.

Both hypotheses of Step 4 hold (with this A_n, B_n, A), so Step 4 gives:

$$\begin{aligned} &A_n + B_n \xrightarrow{d} A \\ \Longrightarrow\ & cX_n + X_n(Y_n - c) \xrightarrow{d} cX. \end{aligned}$$

Since $X_nY_n = cX_n + X_n(Y_n - c)$ (stated at the start of the proof, see (1)) for every $n \in \mathbb{N}$, this is equivalent to:

$$X_nY_n \;\xrightarrow{d}\; cX.$$

□

Part a) ii): $\dfrac{X_n}{Y_n}\mathbb{I}_{\{Y_n\neq 0\}} \xrightarrow{d} \dfrac{X}{c}$ for $c \neq 0$

We deduce this from Part a) i) by first showing that the reciprocal sequence $1/Y_n$ (truncated on $\{Y_n = 0\}$) converges in probability to $1/c$, and then applying Part a) i) to the pair $(X_n, 1/Y_n)$.

Step 1 ($W_n := \dfrac{1}{Y_n}\mathbb{I}_{\{Y_n\neq 0\}} \xrightarrow{\mathbb{P}} \dfrac{1}{c}$). Let $\eta > 0$ be arbitrary, and set $\rho := |c|/2 > 0$ (well-defined since $c \neq 0$ by assumption). For any $\omega \in \Omega$ with $|Y_n(\omega) - c| \leq \rho$, the reverse triangle inequality gives:

$$|Y_n(\omega)| \;\geq\; |c| - |Y_n(\omega) - c| \;\geq\; |c| - \rho \;=\; |c| - \frac{|c|}{2} \;=\; \frac{|c|}{2} \;=\; \rho \;>\; 0.$$

In particular $Y_n(\omega) \neq 0$, so on this event $W_n(\omega) = 1/Y_n(\omega)$ by definition of W_n (rather than 0). We then compute:

$$\begin{aligned} \left|W_n(\omega) - \frac{1}{c}\right| &= \left|\frac{1}{Y_n(\omega)} - \frac{1}{c}\right| \\ &= \left|\frac{c - Y_n(\omega)}{Y_n(\omega)\,c}\right| \\ &= \frac{|Y_n(\omega) - c|}{|Y_n(\omega)|\cdot|c|} \\ &\leq \frac{|Y_n(\omega) - c|}{\rho\cdot|c|} && \text{(since } |Y_n(\omega)| \geq \rho\text{, shown above)} \\ &= \frac{|Y_n(\omega) - c|}{(|c|/2)\cdot|c|} \\ &= \frac{2\,|Y_n(\omega) - c|}{c^2}. \end{aligned}$$

We want the right-hand side to be at most η. Solving for $|Y_n(\omega) - c|$:

$$\frac{2\,|Y_n(\omega) - c|}{c^2} \;\le\; \eta \quad\iff\quad |Y_n(\omega) - c| \;\le\; \frac{\eta c^2}{2}.$$

Therefore, whenever $\omega \in \Omega$ satisfies $|Y_n(\omega) - c| \le \rho$ and $|Y_n(\omega) - c| \le \dfrac{\eta c^2}{2}$, we get:

$$\left|W_n(\omega) - \frac{1}{c}\right| \;\le\; \frac{2\,|Y_n(\omega) - c|}{c^2} \;\le\; \eta.$$

Set $\delta := \min\left(\rho,\ \dfrac{\eta c^2}{2}\right) > 0$. Since $\delta \le \rho$ and $\delta \le \eta c^2/2$, the single condition $|Y_n(\omega) - c| \le \delta$ implies both conditions used above. Therefore, whenever $\omega \in \Omega$ satisfies $|Y_n(\omega) - c| \le \delta$, we get $\left|W_n(\omega) - \frac{1}{c}\right| \le \eta$. In set notation, this is equivalent to:

$$\begin{aligned}
&\{|Y_n - c| \le \delta\} \;\subseteq\; \left\{\left|W_n - \frac{1}{c}\right| \le \eta\right\} \\
\implies\; &\left\{\left|W_n - \frac{1}{c}\right| \le \eta\right\}^c \;\subseteq\; \{|Y_n - c| \le \delta\}^c \qquad\qquad \text{(taking complements)} \\
\implies\; &\left\{\left|W_n - \frac{1}{c}\right| > \eta\right\} \;\subseteq\; \{|Y_n - c| > \delta\}.
\end{aligned}$$

By monotonicity of $\mathbb{P}$ (Lemma 1.1.10d):

$$\mathbb{P}\left(\left|W_n - \frac{1}{c}\right| > \eta\right) \;\le\; \mathbb{P}(|Y_n - c| > \delta).$$

Since $Y_n \xrightarrow{\mathbb{P}} c$, Definition 4.1.1 (applied with threshold δ) gives $\mathbb{P}(|Y_n - c| > \delta) \to 0$ as $n \to \infty$, and therefore:

$$\mathbb{P}\left(\left|W_n - \frac{1}{c}\right| > \eta\right) \xrightarrow{n\to\infty} 0.$$

As $\eta > 0$ was arbitrary, Definition 4.1.1 gives:

$$W_n \xrightarrow{\mathbb{P}} \frac{1}{c}.$$

Step 2 (Apply Part a) i)). We check that the two hypotheses of Part a) i) are satisfied for the pair (X_n, W_n) with constant $\frac{1}{c}$ (which is a well-defined finite real number since $c \neq 0$):

- $X_n \xrightarrow{d} X$ holds by assumption.
- $W_n \xrightarrow{\mathbb{P}} \frac{1}{c}$ holds by Step 1.

Both hypotheses hold, so Part a) i) (applied to X_n, W_n, and constant $\frac{1}{c}$ in place of X_n, Y_n, c) gives:

$$X_n W_n \xrightarrow{d} \frac{1}{c} X \;=\; \frac{X}{c}.$$

For any $\omega \in \Omega$ we distinguish two cases:

- **Case $Y_n(\omega) \neq 0$:** then $W_n(\omega) = 1/Y_n(\omega)$ by definition, so

$$X_n(\omega) W_n(\omega) \;=\; X_n(\omega) \cdot \frac{1}{Y_n(\omega)} \;=\; \frac{X_n(\omega)}{Y_n(\omega)} \;=\; \frac{X_n(\omega)}{Y_n(\omega)} \mathbb{I}_{\{Y_n(\omega) \neq 0\}}.$$

- **Case $Y_n(\omega) = 0$:** then $W_n(\omega) = 0$ by definition, so $X_n(\omega)W_n(\omega) = 0$; likewise $\dfrac{X_n(\omega)}{Y_n(\omega)}\mathbb{I}_{\{Y_n(\omega)\neq 0\}} = 0$, since the indicator vanishes.

In both cases, $X_nW_n = \dfrac{X_n}{Y_n}\mathbb{I}_{\{Y_n\neq 0\}}$ holds pointwise for every $\omega \in \Omega$. Therefore:

$$\frac{X_n}{Y_n}\mathbb{I}_{\{Y_n\neq 0\}} \xrightarrow{d} \frac{X}{c}.$$

$\square$

Part b)

Since g is continuous, it is Borel-measurable (Lemma 3.1.8). The map $\omega \mapsto (X_n(\omega), Y_n(\omega))$ is $\mathcal{F}$-$\mathcal{B}(\mathbb{R}^2)$-measurable, so $g(X_n, Y_n)$ is again a random variable; the same applies to $g(0, Y_n)$ and to their difference (Lemma 3.1.9). We decompose $g(X_n, Y_n)$ into:

$$g(X_n, Y_n) = g(0, Y_n) + \big(g(X_n, Y_n) - g(0, Y_n)\big),$$

which holds pointwise for every $\omega \in \Omega$.

Step 1 (Show $\forall\delta > 0\ \exists M > 0 : \mathbb{P}(|Y_n| > M) < \delta, \forall n \in \mathbb{N}$). Since $Y_n \xrightarrow{d} Y$, we apply exactly the same argument used in Part a), Step 1 (with Y_n, Y in place of X_n, X), which shows:

$$\forall\delta > 0\ \exists M > 0 : \quad \mathbb{P}(|Y_n| > M) < \delta \quad \forall n \in \mathbb{N}.$$

Step 2 ($g(0, Y_n) \xrightarrow{d} g(0, Y)$). Define $\tilde{g} : \mathbb{R} \to \mathbb{R}$, $\tilde{g}(y) := g(0, y)$ for $y \in \mathbb{R}$. Since g is continuous on $\mathbb{R}^2$, $\tilde{g}$ is continuous on $\mathbb{R}$ (as the restriction of g to the line $\{0\} \times \mathbb{R}$). Let $h : \mathbb{R} \to \mathbb{R}$ be bounded and continuous. Then $h \circ \tilde{g}$ is bounded (since h is bounded) and continuous (as a composition of the continuous functions $\tilde{g}$ and h). Since $Y_n \xrightarrow{d} Y$, Definition 4.1.4(ii) applied to $h \circ \tilde{g}$ gives:

$$\begin{aligned}
&\mathbb{E}[(h \circ \tilde{g})(Y_n)] \xrightarrow{n\to\infty} \mathbb{E}[(h \circ \tilde{g})(Y)] \\
\Longrightarrow\ &\mathbb{E}[h(g(0, Y_n))] \xrightarrow{n\to\infty} \mathbb{E}[h(g(0, Y))].
\end{aligned}$$

As h was an arbitrary bounded continuous function, then, by Definition 4.1.4(ii), we have:

$$g(0, Y_n) \xrightarrow{d} g(0, Y).$$

Step 3 ($g(X_n, Y_n) - g(0, Y_n) \xrightarrow{\mathbb{P}} 0$). Let $\varepsilon > 0$ and let $\delta > 0$ be arbitrary. By Step 1, choose $M = M(\delta) > 0$ such that $\mathbb{P}(|Y_n| > M) < \delta$ for all $n \in \mathbb{N}$.

Consider the compact set $K := [-1, 1] \times [-M, M] \subset \mathbb{R}^2$. Since g is continuous on $\mathbb{R}^2$, its restriction to K is continuous on a compact set, hence uniformly continuous on K (Heine–Cantor theorem, known from analysis). Applying uniform continuity to the given $\varepsilon > 0$, there exists $\rho = \rho(\varepsilon, M) \in (0, 1]$ such that:

$$\forall (x_1, y), (x_2, y) \in K, \quad |x_1 - x_2| < \rho \quad \Longrightarrow \quad |g(x_1, y) - g(x_2, y)| < \varepsilon$$

(we may always shrink ρ so that $\rho \leq 1$, since uniform continuity remains valid for any smaller ρ).

For any $\omega \in \Omega$ with $|X_n(\omega)| < \rho$ and $|Y_n(\omega)| \leq M$: since $\rho \leq 1$, both $X_n(\omega) \in [-1, 1]$ and $0 \in [-1, 1]$, and $Y_n(\omega) \in [-M, M]$, so $(X_n(\omega), Y_n(\omega)), (0, Y_n(\omega)) \in K$. Moreover $|X_n(\omega) - 0| = |X_n(\omega)| < \rho$. Applying uniform continuity with $y = Y_n(\omega)$, $x_1 = X_n(\omega)$, $x_2 = 0$:

$$|g(X_n(\omega), Y_n(\omega)) - g(0, Y_n(\omega))| < \varepsilon.$$

Therefore, whenever $\omega \in \Omega$ satisfies $|X_n(\omega)| < \rho$ and $|Y_n(\omega)| \le M$, we get $|g(X_n(\omega), Y_n(\omega)) - g(0, Y_n(\omega))| < \varepsilon$. In set notation, this is equivalent to:

$$\begin{aligned}
& \{|X_n| < \rho\} \cap \{|Y_n| \le M\} \ \subseteq \ \{|g(X_n, Y_n) - g(0, Y_n)| < \varepsilon\} \\
\Longrightarrow \ & \{|g(X_n, Y_n) - g(0, Y_n)| < \varepsilon\}^c \ \subseteq \ \big(\{|X_n| < \rho\} \cap \{|Y_n| \le M\}\big)^c \quad \text{(taking complements)} \\
\Longrightarrow \ & \{|g(X_n, Y_n) - g(0, Y_n)| \ge \varepsilon\} \ \subseteq \ \{|X_n| \ge \rho\} \cup \{|Y_n| > M\} \qquad \text{(De Morgan's law)}
\end{aligned}$$

By sub-additivity (Lemma 1.1.10e), we have:

$$\begin{aligned}
\mathbb{P}\big(|g(X_n, Y_n) - g(0, Y_n)| \ge \varepsilon\big) \ &\le \ \mathbb{P}(|X_n| \ge \rho) + \mathbb{P}(|Y_n| > M) \\
&< \ \mathbb{P}(|X_n| \ge \rho) + \delta \quad \text{(by Step 1).}
\end{aligned}$$

Since $\{|X_n| \ge \rho\} \subseteq \{|X_n| > \rho/2\}$ (as $\rho > \rho/2$), monotonicity of $\mathbb{P}$ (Lemma 1.1.10d) gives $\mathbb{P}(|X_n| \ge \rho) \le \mathbb{P}(|X_n| > \rho/2)$, so:

$$\mathbb{P}\big(|g(X_n, Y_n) - g(0, Y_n)| \ge \varepsilon\big) \ < \ \mathbb{P}(|X_n| > \rho/2) + \delta.$$

Taking $\limsup_{n\to\infty}$ on both sides and using $X_n \xrightarrow{\mathbb{P}} 0$, we get:

$$\begin{aligned}
\limsup_{n\to\infty} \mathbb{P}\big(|g(X_n, Y_n) - g(0, Y_n)| \ge \varepsilon\big) \ &\le \ \limsup_{n\to\infty} \Big[\mathbb{P}(|X_n| > \rho/2) + \delta\Big] \\
&= \ \delta + \limsup_{n\to\infty} \mathbb{P}(|X_n| > \rho/2) \\
&= \ \delta + \lim_{n\to\infty} \mathbb{P}(|X_n| > \rho/2) \qquad \text{(since the limit exists)} \\
&= \ \delta + 0 \qquad \text{(since } X_n \xrightarrow{\mathbb{P}} 0\text{, Definition 4.1.1)} \\
&= \ \delta.
\end{aligned}$$

Since $\delta > 0$ was arbitrary, letting $\delta \downarrow 0$ gives

$$\begin{aligned}
& \limsup_{n\to\infty} \mathbb{P}\big(|g(X_n, Y_n) - g(0, Y_n)| \ge \varepsilon\big) \le 0 \\
\Longrightarrow \ & \limsup_{n\to\infty} \mathbb{P}\big(|g(X_n, Y_n) - g(0, Y_n)| \ge \varepsilon\big) = 0 \quad (\mathbb{P} \ge 0).
\end{aligned}$$

Since $\{|g(X_n, Y_n) - g(0, Y_n)| > \varepsilon\} \subseteq \{|g(X_n, Y_n) - g(0, Y_n)| \ge \varepsilon\}$, monotonicity of $\mathbb{P}$ (Lemma 1.1.10d) gives:

$$\lim_{n\to\infty} \mathbb{P}\big(|g(X_n, Y_n) - g(0, Y_n)| > \varepsilon\big) = 0.$$

As $\varepsilon > 0$ was arbitrary, Definition 4.1.1 gives:

$$g(X_n, Y_n) - g(0, Y_n) \xrightarrow{\mathbb{P}} 0.$$

Step 4 ($g(X_n, Y_n) \xrightarrow{d} g(0, Y)$). Define $A_n := g(0, Y_n)$ and $B_n := g(X_n, Y_n) - g(0, Y_n)$ for $n \in \mathbb{N}$. We check that the two hypotheses of Step 4 of Part a) i) are satisfied for this choice:

- By Step 2, $A_n = g(0, Y_n) \xrightarrow{d} g(0, Y)$. We therefore set $A := g(0, Y)$.
- By Step 3, $B_n = g(X_n, Y_n) - g(0, Y_n) \xrightarrow{\mathbb{P}} 0$.

Both hypotheses hold (with this A_n, B_n, A), so Step 4 of Part a) i) gives:

$$\begin{aligned}
& A_n + B_n \xrightarrow{d} A \\
\Longrightarrow \ & g(0, Y_n) + \big(g(X_n, Y_n) - g(0, Y_n)\big) \xrightarrow{d} g(0, Y).
\end{aligned}$$

Since $g(0, Y_n) + \big(g(X_n, Y_n) - g(0, Y_n)\big) = g(X_n, Y_n)$ for every $n \in \mathbb{N}$ (and pointwise), this is equivalent to:

$$g(X_n, Y_n) \ \xrightarrow{d} \ g(0, Y).$$

□